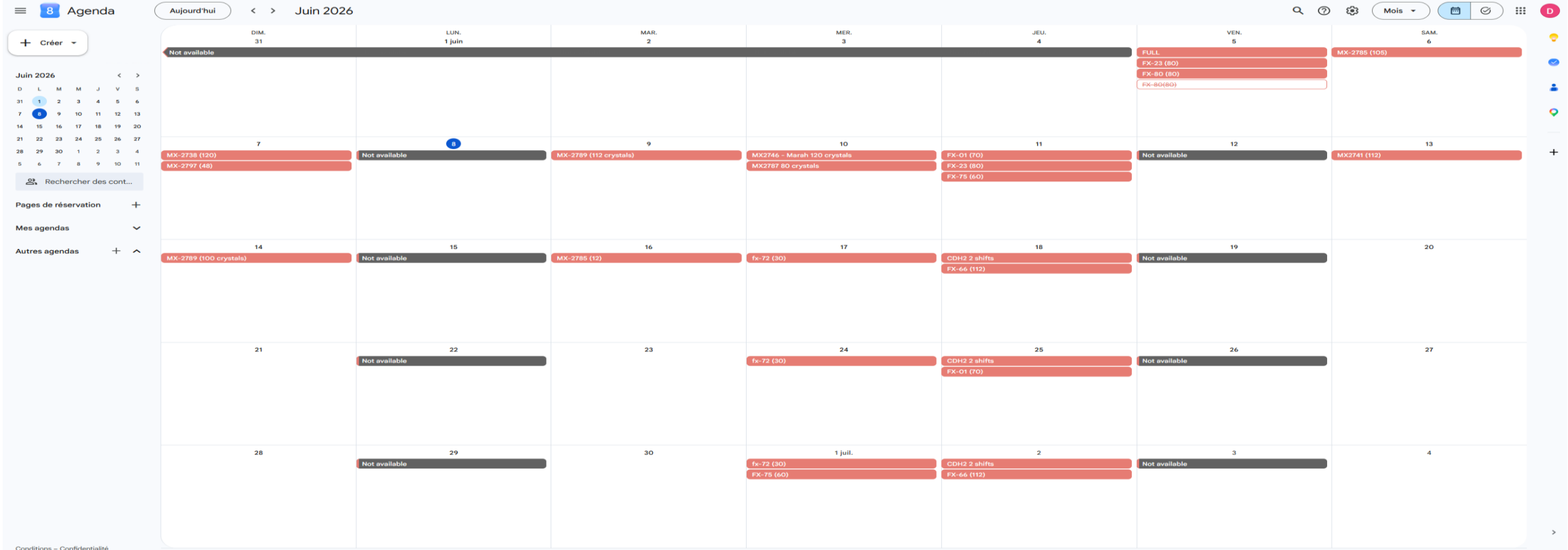




MASSSIF-1, the automated beamline at ESRF

Didier NURIZZO
MXIS Group

Scheduling



- ☐ ESRF works in shift of 8 hours
- ☐ Google calendar is owned by the operation team managing the session scheduling and the invitations
- ☐ Allocated shifts number is calculated per samples with a maximum of 60 samples / shift
- ☐ Number of samples are given for information

Flexibility and User friendliness - Data collection happen in ≤ 3 working days

Data Portal for Parcel

[List](#)[Spreadsheet](#)[+ New parcel](#)

	Name	Items #	Status	Dangerous goods	Reimbursed	After experiment
	Aurora-parcel	4	 SCHEDULED	No	No	Destroy

Items 1-1 of 1

Show 20 ▾

Create parcel

✕

Name*

Description

Storage conditions*

Comments

Save

Cancel

How to create the Parcel

The parcel is linked to a scheduled session

Print out of the stickers with addresses

1 Dewar per shift is covered by ESRF

Data Portal for the Diffraction Plan and the Processing Plan

[List](#)[Spreadsheet](#)[+ New parcel](#)

Name	Items #	Status	Dangerous goods	Reimbursed	After experiment
 Aurora-parcel	4	 SCHEDULED	No	No	Destroy

Items 1-1 of 1

[Show 20](#) ▾

Content

[Containers](#)[Spreadsheet](#)[+ New container](#)

CPS7997



CPS7861



CPS7998



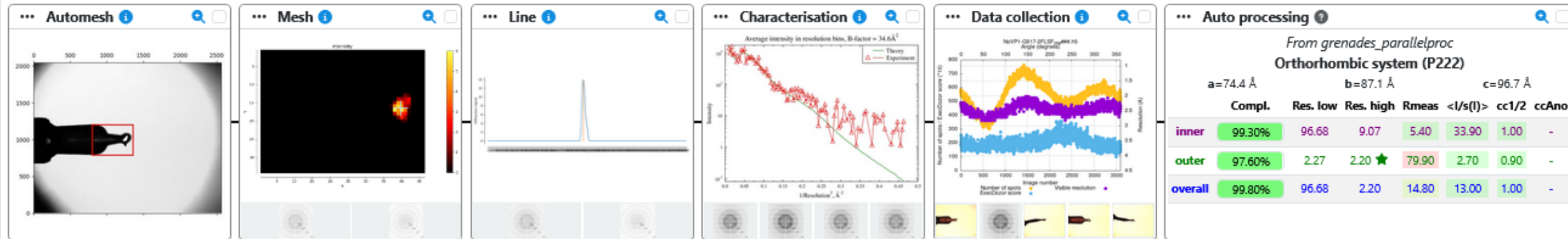
BIOSYS-01

☐ How to create the containers☐ Access to the diffraction plan and the processing plan

- ❖ Experiment type
- ❖ Min rotation, total rotation
- ❖ resolution
- ❖ Beam diameter
- ❖ Radiation sensitivity
- ❖ Number of positions
- ❖ 2nd high dose data set (factor)
- ❖ Unit cell, Completeness, multiplicity
- ❖ MR from cell
- ❖ Models, UniProt ID
- ❖ SMILES
- ❖ Reference reflections

Data Collection Workflow

07/06/2026 15:59:05 MXPressE



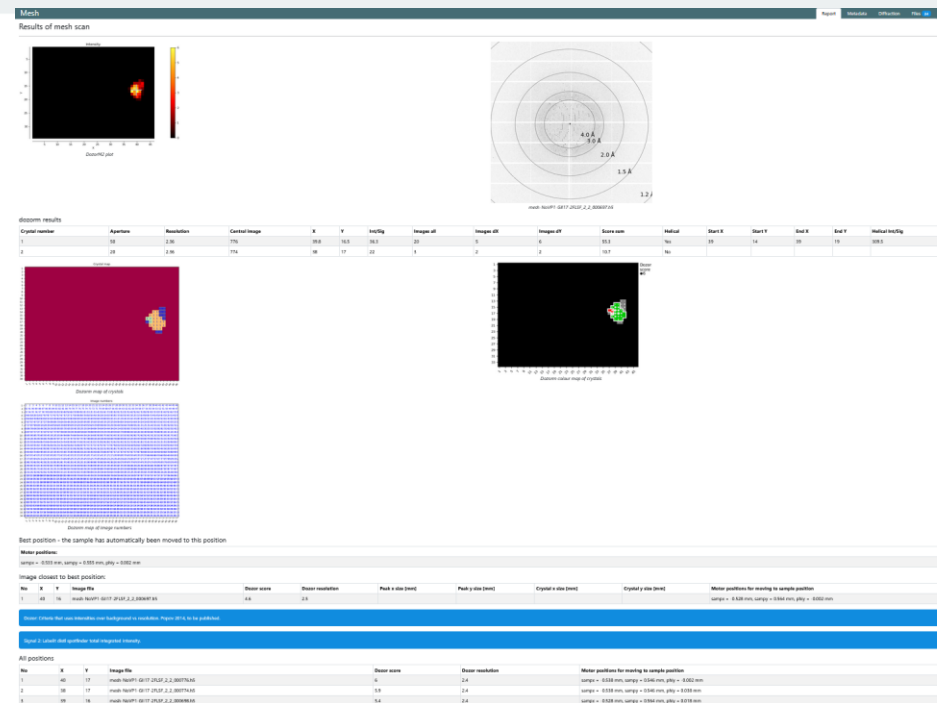
Predefined parameters: total rotation = 360.0 degrees, aimed completeness = 98.0 %, observed resolution = 1.50 Å, DC resolution = 1.50 Å, min osc width = 0.10 degrees; Warning! Cannot set aimed multiplicity as total rotation is forced.; Auto mesh used., Dynamic aperture set to 50 um; Using forced space group 'P21' from diffraction plan. Characterisation: Space group P21 forced., Auto processing (except Grenades parallelproc): Forced space group P21 and cell (a=60.2, b=87.4, c=62.4, alpha=90.0, beta=105.3, gamma=90.0).

Optical mesh (6 images - Lucid)

- ❖ Optical centering is part of the sample loading
- ❖ Calculate the largest surface of the loop
- ❖ Define the ROI

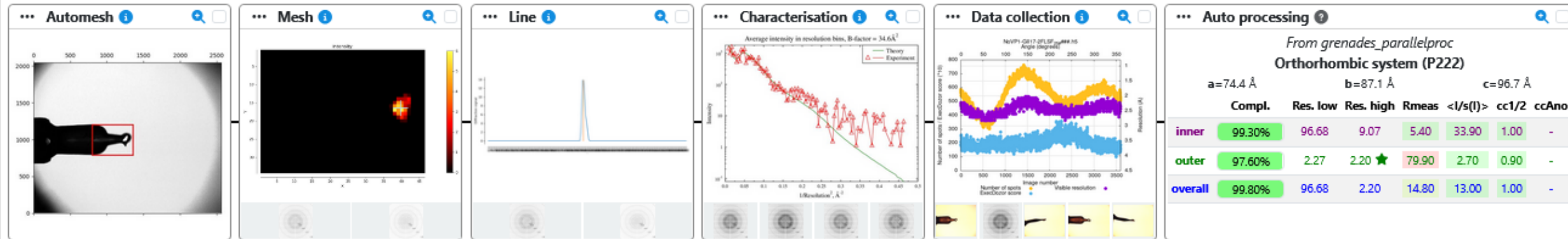
Xray 2D-scan

- ❖ Best diffraction image
- ❖ 2D COM (motor positions)
- ❖ 2D crystal size
- ❖ Isomorphism characterization, Dozor score and resolution limit



Data Collection Workflow

07/06/2026 15:59:05 MXPressE



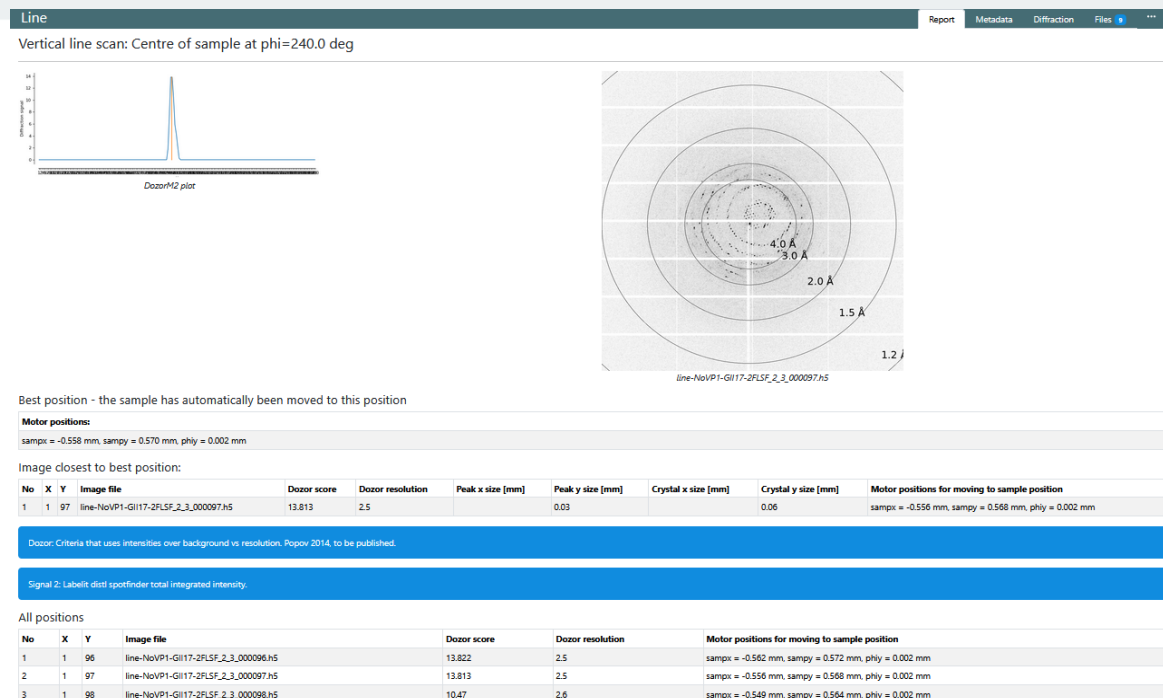
Predefined parameters: total rotation = 360.0 degrees, aimed completeness = 98.0 %, observed resolution = 1.50 Å, DC resolution = 1.50 Å, min osc width = 0.10 degrees; Warning! Cannot set aimed multiplicity as total rotation is forced.; Auto mesh used., Dynamic aperture set to 50 um; Using forced space group 'P21' from diffraction plan. Characterisation: Space group P21 forced., Auto processing (except Grenades parallelproc): Forced space group P21 and cell (a=60.2, b=87.4, c=62.4, alpha=90.0, beta=105.3, gamma=90.0).

Xray 2D-scan

- ❖ Best diffraction image
- ❖ 2D COM (motor positions)
- ❖ 2D crystal size
- ❖ Isomorphism characterization, dozor score and resolution limit

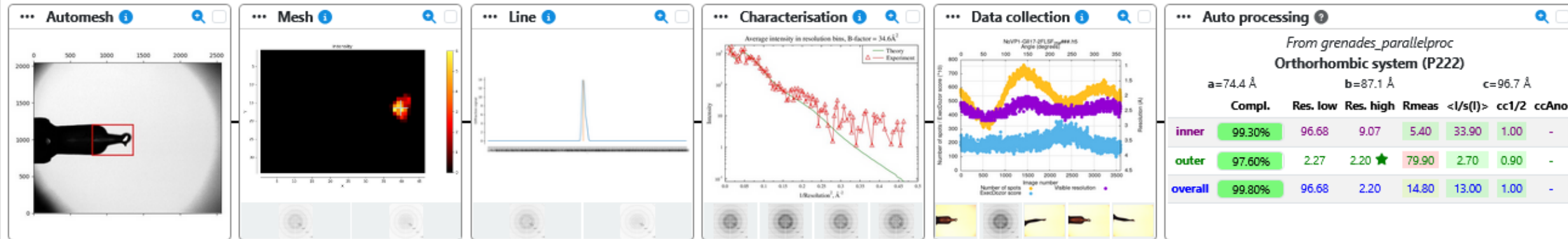
Line scan at 90°

- ❖ Best diffraction image
- ❖ 3D COM (motor positions)
- ❖ 3D crystal size
- ❖ Dozor score and resolution limit



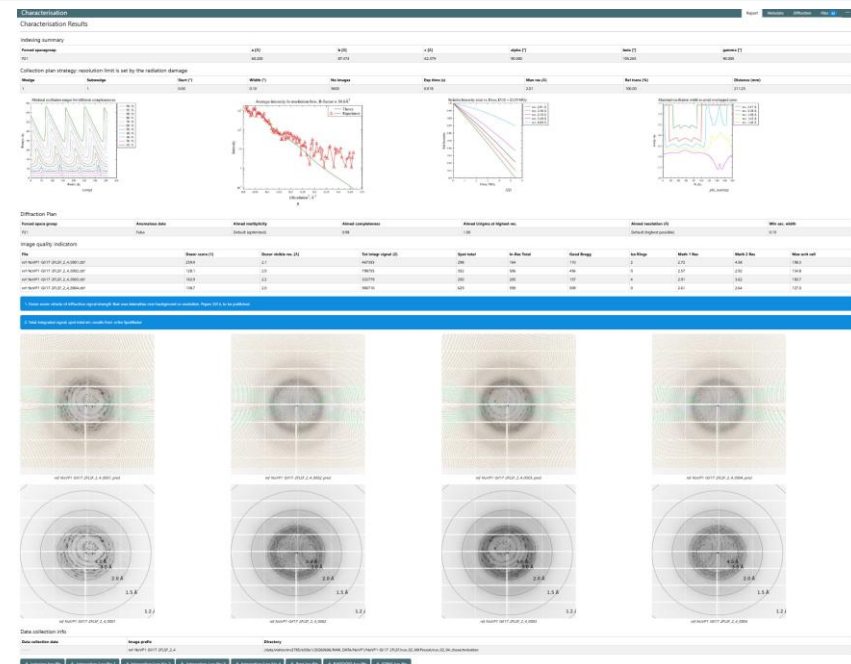
Data Collection Workflow

07/06/2026 15:59:05 MXPressE

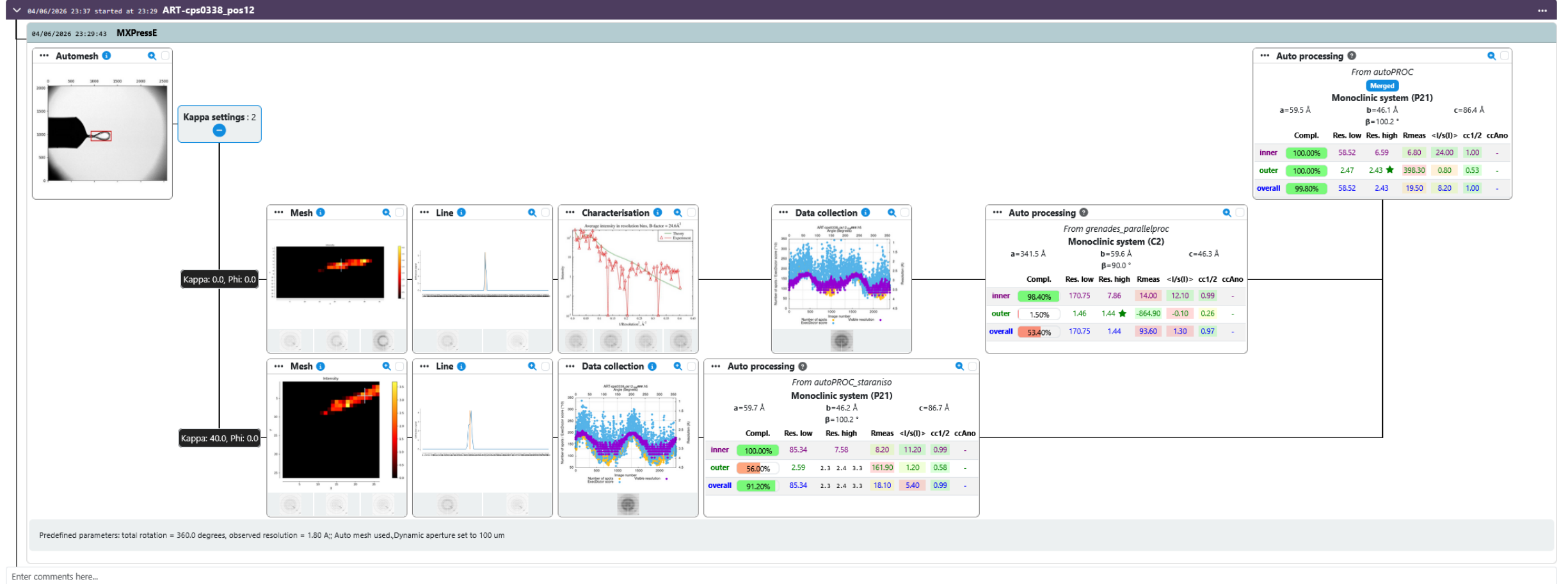


Predefined parameters: total rotation = 360.0 degrees, aimed completeness = 98.0 %, observed resolution = 1.50 Å, DC resolution = 1.50 Å, min osc width = 0.10 degrees; Warning! Cannot set aimed multiplicity as total rotation is forced.; Auto mesh used., Dynamic aperture set to 50 um; Using forced space group 'P21' from diffraction plan. Characterisation: Space group P21 forced., Auto processing (except Grenades parallelproc): Forced space group P21 and cell (a=60.2, b=87.4, c=62.4, alpha=90.0, beta=105.3, gamma=90.0).

- ❑ Characterization (4 images – 10 x 0.1°)
 - ❖ Xray centring control with recentring
 - ❖ Redo characterization depending on resolution
 - ❖ Resolution limit based on line/mesh scan
 - ❖ Total exposure with crystal size
 - ❖ Beam and crystal size tailoring
 - ❖ Space group indexing and strategy
 - ❖ Triclinic and monoclinic Kappa reorientation
 - ❖ User inputs (radiation sensitivity, total rotation, etc...)



Kappa re-orientation



- ❑ For Triclinic and monoclinic lattices with resolution better than 2.0 Å
- ❑ Optical realignment after kappa opening at 40°
- ❑ Xray centering on the new center point
- ❑ Automatic processing and merging of the two data sets

Automatic Data Processing

Auto processing ranking

We use the following criteria by order of priority:

1. Matches all set filter cutoffs

Add...

2. Highest symmetry space group

Enabled

3. Selected criteria

Outer

Res. high

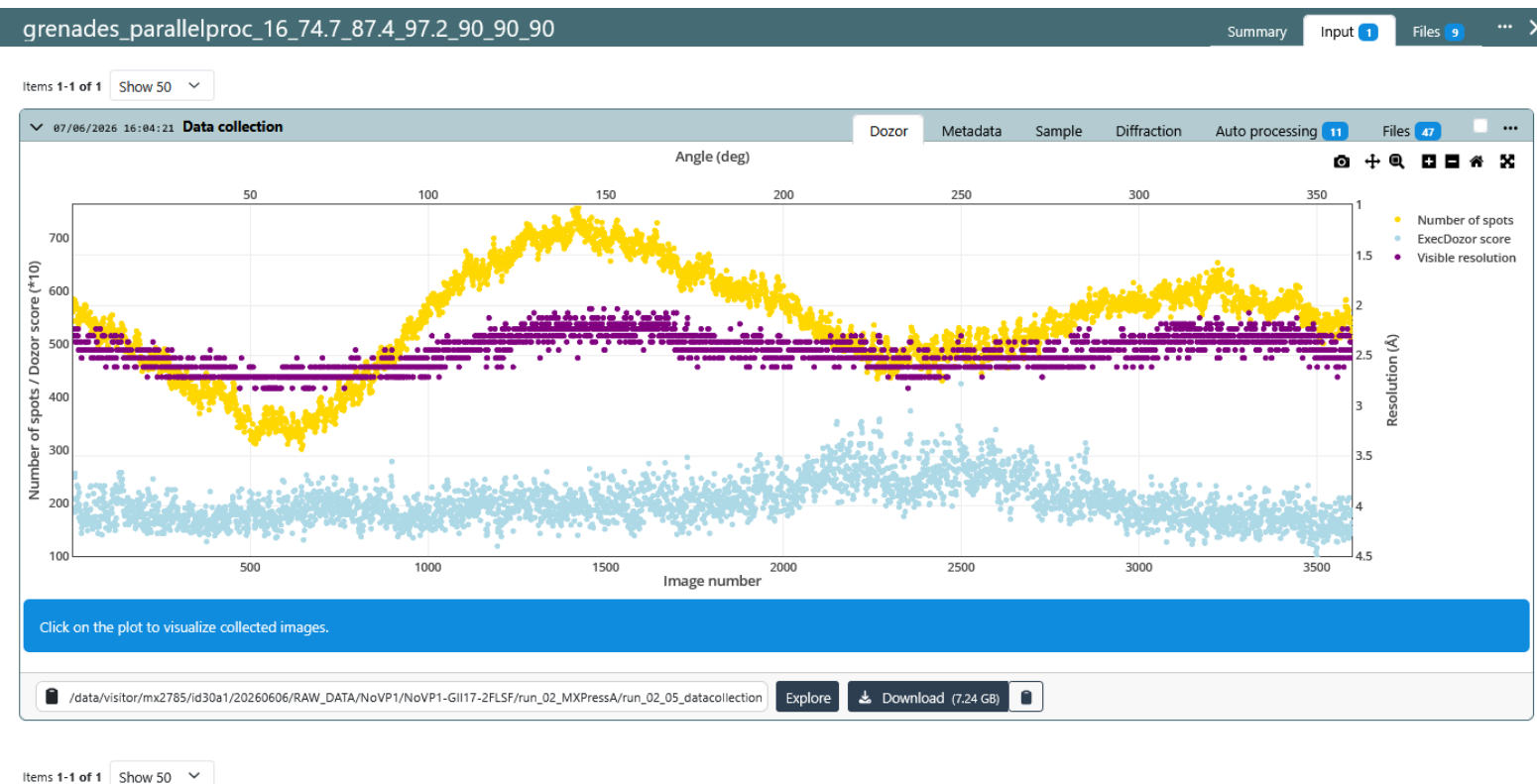
Cutoff Configuration

	Inner	Outer	Overall
Compl. <			
Res. low <			
Res. high >			
Rmeas >			
<I/s(I)> <			
cc1/2 <			
ccAno <			

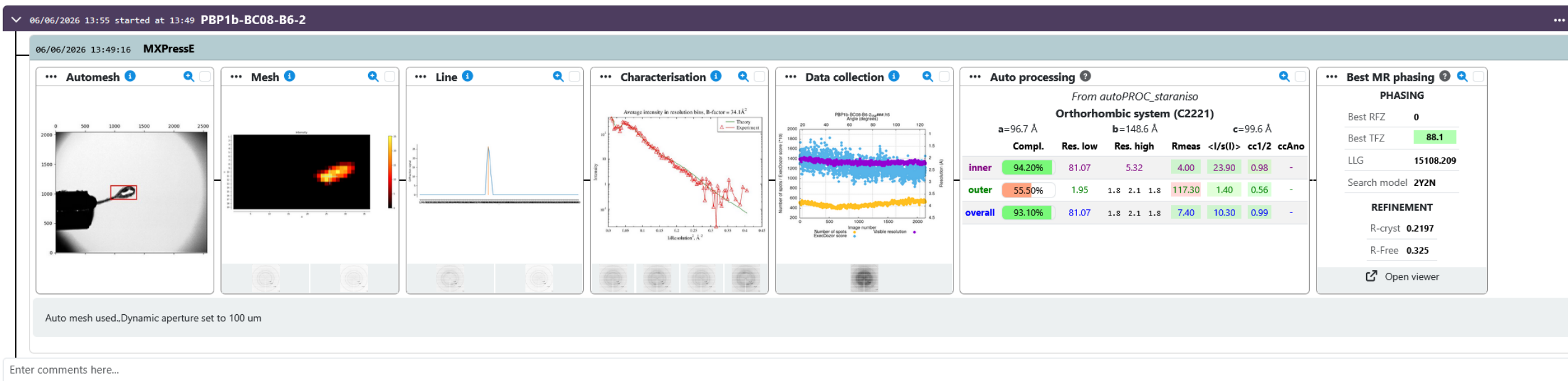
Apply Cancel

- Best processing ranking parameters are selected by the user
- Easy download of the results
- Access to:
 - Dozor scores
 - Metadata
 - Logfiles
 - Diffraction images
 - MTZ files

Auto processing ranking										
	Program	a,b,c (Å)	α,β,γ (°)	Compl.	Res. low	Res. high	Rmeas	<I/s(I)>	cc1/2	ccAno
...	grenades_parallelproc P222	74.4	90.0	inner 99.30%	96.68	9.07	5.40	33.90	1.00	-
		87.1	90.0	outer 97.60%	2.27	2.20	79.90	2.70	0.90	-
		96.7	90.0	overall 99.80%	96.68	2.20	14.80	13.00	1.00	-
...	grenades_parallelproc P2	73.6	90.0	inner 99.30%	96.01	5.53	8.70	20.70	0.99	-
		86.3	90.1	outer 0.00%	1.03	1.01	0.00	0.10	0.00	-
		96.0	90.0	overall 44.30%	96.01	1.01	47.40	2.20	0.98	-
...	autoPROC_staraniso P21	60.9	90.0	inner 99.70%	58.91	5.48	6.80	12.60	0.99	-
		87.1	104.9	outer 46.10%	1.83	1.8	111.40	1.50	0.60	-
		61.0	90.0	overall 74.00%	58.91	1.8	17.00	5.80	0.99	-
...	XIA2_DIALS P21	60.7	90.0	inner 99.77%	58.67	4.58	7.07	30.43	0.98	-
		86.7	105.0	outer 91.35%	1.72	1.69	337.08	0.19	0.13	-
		60.6	90.0	overall 97.74%	58.64	1.69	19.69	3.76	0.99	-
...	autoPROC P21	60.9	90.0	inner 99.70%	58.91	4.75	7.20	12.30	0.99	-
		87.1	104.9	outer 97.00%	1.78	1.75	317.10	0.50	0.43	-
		61.0	90.0	overall 98.20%	58.91	1.75	22.20	4.00	0.99	-
...	grenades_parallelproc P2	86.8	90.0	inner 99.00%	96.34	9.85	5.20	26.00	1.00	-
		74.1	90.0	outer 94.60%	2.32	2.26	72.50	2.10	0.86	-
		96.3	90.0	overall 99.50%	96.34	2.26	14.20	9.90	1.00	-
...	grenades_parallelproc P2	74.3	90.0	inner 99.30%	96.60	9.83	6.60	26.00	0.95	-
		96.6	90.0	outer 77.70%	2.32	2.26	72.50	2.00	0.85	-
		87.1	90.0	overall 98.20%	96.60	2.26	14.40	9.90	0.99	-
...	grenades_parallelproc P1	69.8	97.8	inner 98.30%	98.56	9.85	147.80	2.40	0.64	-
		82.1	90.7	outer 6.50%	3.12	2.97	39.60	1.50	0.81	-
		99.5	90.2	overall 77.20%	98.56	2.97	75.30	2.30	0.69	-
...	grenades_parallelproc P1	70.2	82.3	inner 98.40%	98.93	9.89	140.60	2.50	0.57	-
		82.6	89.2	outer 8.90%	3.13	2.98	36.60	1.50	0.84	-
		98.8	90.1	overall 78.90%	98.93	2.98	72.40	2.30	0.69	-
...	grenades_parallelproc P1	69.7	97.4	inner 98.20%	97.45	10.02	172.50	2.40	0.45	-
		82.0	90.8	outer 18.20%	3.17	3.02	38.50	1.70	0.81	-
		98.3	90.2	overall 81.20%	97.45	3.02	76.50	2.30	0.67	-
...	grenades_parallelproc P1	70.7	96.3	inner 98.20%	97.98	10.25	224.20	2.40	0.24	-
		83.1	89.4	outer 26.10%	3.24	3.09	38.40	1.80	0.74	-
		98.6	89.9	overall 84.80%	97.98	3.09	79.80	2.30	0.51	-

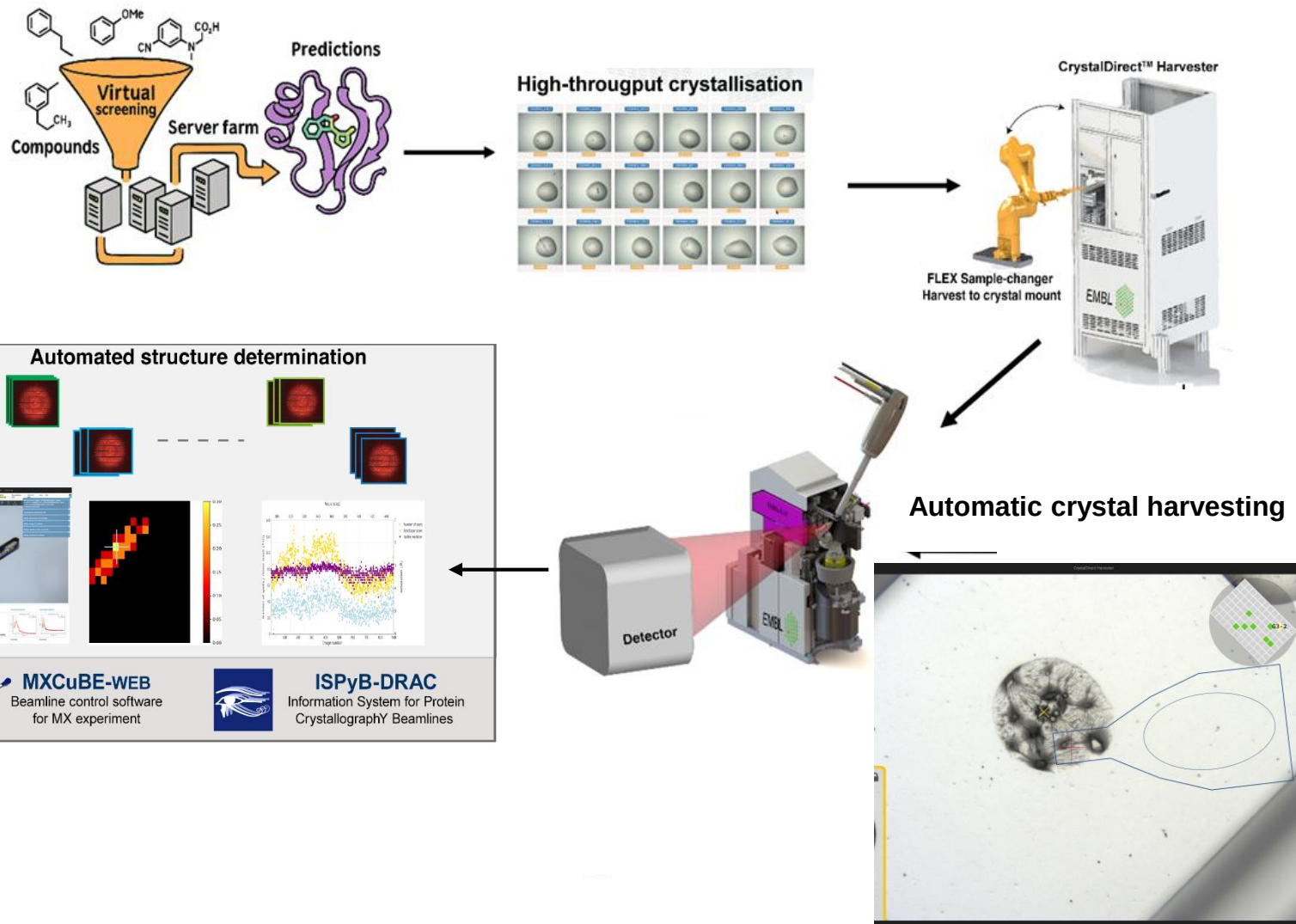


Molecular Replacement and Compound Fitting



- ☐ The MR is based on:
 - ❖ Cell dimension
 - ❖ Model provided by user with access to RCSB PDB
- ☐ TFZ and LLG score given for the best models
- ☐ Automatic refinement of the MR structure
- ☐ On-line access to Moorhen for structure and density visualization
- ☐ Compound fitting in coot

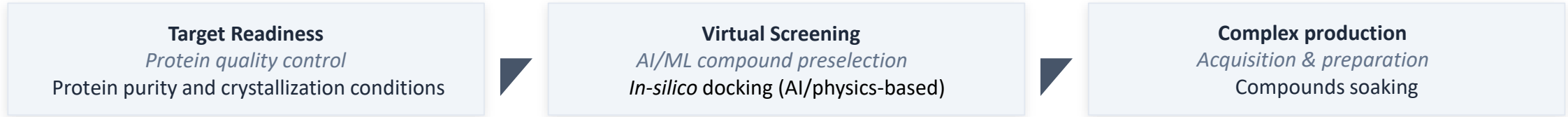
Room Temperature Compound Screening



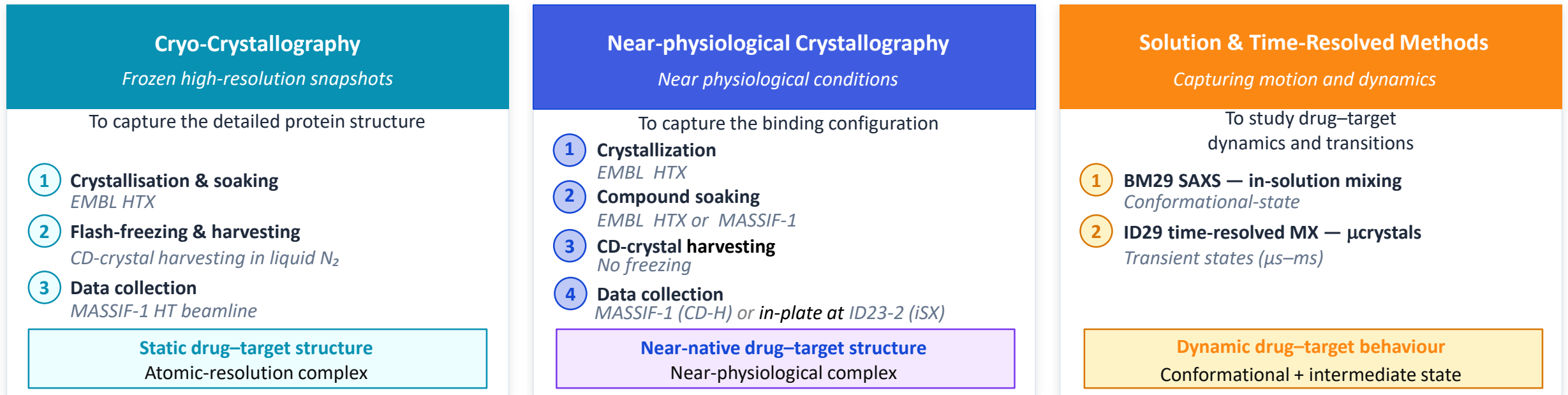
- ☐ Virtual compound screening
- ☐ HTX crystallization
- ☐ Compound soaking:
 - ❖ Drop dispenser (Mosquito/Echo)
 - ❖ TouchTip
- ☐ Room temperature harvesting
- ☐ Data collection in humidity controlled environment
- ☐ Automatic data processing

Overview of the Automated Near-native Compound Screening Pipeline

Phase I: preparation



Phase II: structural analysis, three complementary approaches



- An integrated, iterative and high throughput user service
- Computational pre-filtering of the compound libraries
- Rapid structure determination at different temperatures



Macromolecular X-ray Crystallography and Imaging group

**Beamline Control Unit (Bliss/MXCuBE)
Data Automation Unit (WF)
Algorithms and scientific Data Analysis group**

**Arinax
User Office – Technical Infrastructure Division
Business Development Office – Safety**

EMBL



**Instrumentation team
HTX-lab team
Synchrotron crystallography team**



THE **P**ARTNERSHIP FOR **S**TRUCTURAL **B**IOLOGY

**ADVANCED METHODS for AMBIENT
CRYSTALLOGRAPHY Workshop**

AMAX2026 – 02-04 December 2026



THANK YOU FOR YOUR ATTENTION

MXCuBE Automation Working Group

- ❑ Launched in Trieste in May 2024
- ❑ 7 meetings from September 2024 to April 2026
- ❑ To have a Long-term perspective on the DC model for automation
- ❑ Minutes are in:

https://github.com/mxcube/mxcube.github.io/tree/master/doc/developers_meetings/Minutes_AutomationWG_Date.pdf

- Focus on X-ray centering

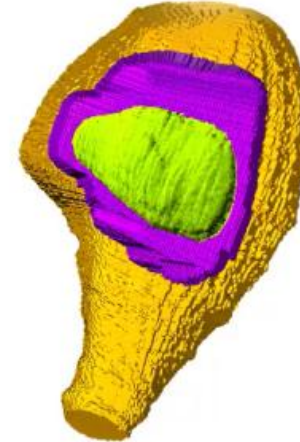
- Input :

- ❖ Sample parameters

- Implementation of the “volume of interest” in MXLIMS
- X-ray sensitivity

- ❖ Beamline and Acquisition parameters

- Same as the one that currently used for a standard DC
- Default value to be defined per BL



Yellow: Volume of the loop
Purple: diffracting volume
Green: best diffracting volume

- ❑ VolumeScan in MXLIMS
 - ❖ Experiment type (String)
 - ❖ Rotation angles (float list)
 - ❖ Results of the search volume with sub-volumes for big scans and micro-beam (PointCloud)
 - ❖ Shape of the box (String)

- ❑ PointCloud
 - ❖ Best position as a dictionary of string and float for motor positions
 - ❖ Grouping type (0=bkg, 1=Xtal1, 2=Xtal2, ...)
 - ❖ Quality indicator (list of float)

- ❑ Introduction of the concept of quality indicator in MXCuBE and how to display a volume with quality indicators
- ❑ Option of showing different type of heat map
- ❑ The results follows the new abstract diffractometer
- ❑ PR to be discussed
- ❑ Strong link with UI/UX WG
 - ❖ Need of sharing (Heat map display)
 - ❖ Meeting once a month UI/UX – Automation

Thank you to all participants